LAST 5 YEARS QUESTIONS

Class : B.Tech I^{st} yr II SEM

SUBJECT : ENGG. MATHEMATICS-II

UNIT → 1

Q.1: State the Cayley Hamilton Theorem. [RTU 2023] & 2022

Q.2: Determine the Rank of matrix [RTU 2023]

Q.3: Verify Cayley Hamilton Theorem for the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Hence find A^{-1}.

[RTU 2023, RTU 2022]

Q.4 Reduce the matrix in Its normal form and hence find Its Rank. [RTU 2023]

Q.5: Define symmetric and skew symmetric matrices. [RTU 2021] & ~~RTU 22~~

Q.6: Define Echelon form of matrix. [RTU 2021]

Q.7: Find the Eigen values and Eigen vectors.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

[RTU 2020,21]

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Q.8: State Cayley Hamilton Theorem. Verify It for the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ and find A^{-1}. [RTU 2023]

Q.9: Find the Eigen value and Eigen vectors of the following matrix [RTU 2019]

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Q.10 State Cayley Hamilton Theorem, Verify It for the following matrix [RTU 2019]

$$\begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

Q.11 State the Rank nullity Theorem. [RTU 2019]

Q.12 Examine whether the following matrix A is Diagonalizable. If so, obtain the matrix P such that $P^{-1}AP$ is a diagonal matrix. [RTU 2019]

Q.13 Let V_1, V_2, V_3 be first, second, third column vectors respectively of matrix $A = \begin{pmatrix} 2 & 1 & 7 \\ 1 & 0 & 2 \\ -1 & 5 & 13 \end{pmatrix}$ what can we Conclude about rank(A) from the observation $2V_1 + 3V_2 - V_3 = 0$ [RTU 22]

Q.14 Examine whether the matrix

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

is diagonalizable. If so, obtain the matrix that $P^{-1}AP$ is a digonal matrix [RTU 22]

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